

Harmful Manipulation and Election Security

The Capability-Deployment-Effect Gap

How frontier AI can turn political influence into infrastructure, and how democracy can be more resilient to the new challenges ahead.

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WORKING-PAPER STATUS

Status. This programme-level AI governance paper integrates operational evidence, capability measurement, election records, and a source-linked policy review. It is not peer-reviewed. Parts I–III retain their 28 August 2026 evidence freeze. Part IV incorporates selected external research reviewed on 1 September 2026; these findings are not new fellowship experiments or automatic upgrades to the production Atlas’s empirical flags.

Claim boundary. The programme’s technical and govtech artefacts do not establish a successful intervention, live audience exposure, human persuasion, or an AI-attributable electoral outcome. External studies support bounded human or technical effects, not end-to-end electoral protection or durable preservation of democratic agency.

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Abstract

Frontier AI does not need to persuade an electorate to create a democratic governance problem. The risk emerges when model capability is joined to audience data, trusted identity, tools, distribution, feedback, and control of the evidence. Influence can then become adaptive infrastructure: persistent, personalised, difficult to observe, and controlled by actors able both to shape the process and to determine what outsiders can know about it.

This paper develops the Capability-Deployment-Effect Gap, or CDE Gap, to separate model behaviour, served-system configuration, deployment, authentic exposure, human and institutional response, and electoral consequence. It uses agency transfer as the democratic severity test: whether practical control over attention, preference formation, choice, dependency, or institutional decisions shifts away from citizens and public institutions towards the actor controlling the system.

Four linked research strands provide a bounded evidence base. The technical work shows how unstable access, routes, parsers, evaluators, and missingness constrain apparently precise capability claims. Election records document operations and institutional responses more readily than authentic exposure or effect. A source-linked policy review finds usable human and technical evidence, but also divergence between recognition, trust, sharing, and choice. Controls that work after delivery may arrive too late; controls that secure an agent may protect its operator rather than the people affected. The policy gap is therefore not simply a shortage of interventions. It also concerns mismatched outcomes, incomplete delivery, and misallocated authority.

The governance implication is to match intervention to both mechanism and beneficiary. Use bounded controls where the relevant endpoint has been tested; make consequential authority inspectable and revocable; test practical exit; and scrutinise defensive monitors as well as the systems they monitor. These are differentiated recommendations, not a ranking of proven policies. The democratic objective is neither maximum resistance to AI advice nor merely fewer false messages. It is the continuing ability of citizens and institutions to understand, direct, refuse, contest, and leave the systems through which decisions are made.

The argument in brief

"you shall know the truth and the truth shall make you free"

- John 8:32

The next Cambridge Analytica may not look like a scandal. It may look like infrastructure: a model connected to political data, trusted identity, distribution, feedback, and analytics, each ordinary in isolation and consequential in combination.

Frontier AI may matter less by transforming the average persuasion effect than by changing the architecture of influence. A controller can generate, personalise, translate, test, distribute, observe, and adapt through the same technical stack, while outsiders see only fragments.

The CDE Gap turns that concern into a standard of proof. Capability is not deployment; delivery is not authentic exposure; exposure is not durable effect. Yet institutions need not wait for a national vote-effect estimate before addressing an evidenced mechanism such as impersonation, unlawful targeting, unbounded permissions, opaque distribution, or failed evidence custody.

The democratic objective is not to eliminate political influence. It is to preserve self-correction. The same discipline must apply to defences: whose decision is improved, whose authority is secured, and can the affected person challenge or leave the protective system? A safer agent is not necessarily a less powerful operator.

Table 1. Four research lenses on one influence system. Programme observations, external findings, and analytical inferences retain separate claim ceilings. Source: author synthesis of Guerrero 2026a–f and the cited research.

Research lens	What the programme observed	Governance lesson	Claim ceiling
I. Operations	Model access and served-system traces; intervention conversion can fail, while route outputs may still contain campaign-planning assistance after tactical refusals.	Govern exact routes, configurations, permissions, persistence, and evidence custody—not model names alone.	No successful intervention, autonomous execution, authentic audience contact, persuasion, or electoral effect.
II. Evaluation	Incomparable constructs and pipeline missingness can make benchmark precision misleading.	Treat provider route, parser, simulated interlocutor, evaluator, judge, and missingness as part of the measured system.	No model ranking, longitudinal trend, human-effect estimate, or election effect.
III. Field	Operations, distribution events, attribution evidence, and institutional responses are documented more often than authentic exposure or effect.	Preserve delivery denominators and separate occurrence, attribution, exposure, response, and electoral consequence.	No prevalence estimate, causal zero-effect claim, durable agency-transfer estimate, or AI-attributable national-election outcome.
IV. Policy	External evidence distinguishes human and technical outcomes, delivery constraints, and whose authority a control protects.	Evaluate the measured endpoint, coverage, timing, and beneficiary. Preserve meaningful refusal, revision, and functional exit.	No universal efficacy ranking or demonstrated durable agency preservation. External findings are not fellowship experiments.

The paper uses five evidence labels. Established evidence is directly supported by the cited record. Strong inference joins established facts through a well-supported mechanism but remains unobserved at the endpoint. Plausible hypothesis identifies a testable pathway with material missing links. Speculative scenario describes a consequential possibility whose joined conditions have not been shown. Open question marks what the present design or record cannot resolve.

The Guerrero (2026a–f) citations identify primary materials produced or curated within this research programme. They provide traceability to protocols, records, recovered data, and coding decisions; they are not independent corroboration or peer review. Independent research, public records, provider documentation, and programme materials are separated in the reference list.

1. The next Cambridge Analytica may look ordinary

Cambridge Analytica is remembered as a story about psychographic persuasion. The official record is less cinematic and more important. Regulators established deceptive data collection, voter profiling, opaque targeting, and serious failures of privacy and accountability. They did not establish a clean estimate of votes changed. The durable lesson was architectural: a private actor assembled data, political purpose, audience selection, content, distribution, and measurement before citizens or institutions could inspect the system in time (Information Commissioner's Office 2018; Federal Trade Commission 2019).

The next version may not be a notorious consultancy. It may look like an ordinary stack: an audience database, a model, an assistant, a political-advertising account, a messaging channel, and an analytics dashboard. Each component may be familiar. The democratic significance lies in how they are joined, who controls those joins, and whether anyone outside the system can reconstruct what happened before the decision window closes.

Frontier AI changes this architecture not because every model output is more persuasive than every human message, but because generation, personalisation, translation, testing, and adaptation can be connected cheaply and persistently. The same system can remain present across repeated exchanges, observe response, and try again.

The most consequential shift may therefore be operational rather than rhetorical. Integrated systems may reduce the expertise and coordination needed to assemble an influence operation, while persistent interfaces create repeated opportunities to learn about a person or institution. The relevant object is the deployed system—not the isolated message.

This research does not observe a closed loop operating against real people. That remains a plausible hypothesis. But the possibility changes the question election-security research must ask.

The question is not only whether AI can persuade. It is who can join capability, data, identity, distribution, feedback, and evidence—and what power follows.

Two errors dominate the debate. The alarmist error begins with an impressive output and ends with an altered election. The dismissive error begins with the absence of a vote-effect estimate and concludes that nothing important happened. Both erase the operating system in the middle.

A model output is not a campaign. A campaign is not authentic exposure. Exposure is not persuasion. Yet impersonation, covert targeting, unlawful data use, excessive action authority, or failed evidence preservation do not become harmless because the final electoral effect remains unknown. The CDE Gap is designed to hold both propositions at once.

2. Influence, manipulation, and agency transfer

Manipulation carries too much conceptual weight to describe every form of political influence. Influence is the wider category. Analysis should then separate three questions: what the actor intended, which technique the system used, and what happened to the target. Those questions can come apart. A manipulative attempt can fail. A true claim can be delivered through impersonation. A system can create dependency without changing a political belief.

Persuasion is part of democracy. Political actors offer arguments, evidence, values, promises, and interpretations. Citizens accept some, reject others, and change their minds. Persuasion preserves reflective agency when the source and purpose are reasonably visible, claims can be examined, alternatives remain accessible, and refusal is practical.

Manipulation weakens those conditions. It may rely on deception, impersonation, concealed purpose, vulnerability exploitation, asymmetric personalisation, emotional coercion, dependency, or control over the decision environment. The concern is not merely whether a target changes their mind. It is whether the process impairs reflective and contestable choice (Susser, Roessler, and Nissenbaum 2019; El-Sayed et al. 2024).

The EU General-Purpose AI Code of Practice now supplies an operational regulatory analogue. Its Safety and Security chapter treats harmful manipulation as a specified systemic risk involving strategic distortion of behaviour or beliefs through persuasion, deception, or personalised targeting, especially in multi-turn and difficult-to-detect interactions. That is useful for provider risk management under Article 55 (European Commission 2025b; European Commission 2026).

The Code is not a universal legal definition, a standalone prohibition, or evidence that manipulation occurred. This paper retains the stricter normative question: did the influence process exploit opacity, vulnerability, asymmetry, or dependency in a way that impaired reflective and contestable choice?

The downstream democratic mechanism is agency transfer: practical control over attention, preference formation, choice, dependency, or institutional decision-making shifts away from an affected person, public, or democratic institution and towards the actor controlling the system.

Not every delegation to AI is agency transfer. People routinely delegate tasks while retaining meaningful control. The relevant questions are whether they can understand what the system is doing, identify the principal behind it, refuse it, contest it, correct it, leave it, and recover from its decisions.

Agency transfer can occur at four levels. At the micro level, a system may exploit private data, emotional state, trust, or cognitive limits. At the meso level, assistants, recommenders, and agents may become indispensable intermediaries. At the macro level, dominant interfaces may gain agenda-setting power, epistemic authority, or privileged access to institutions, extending wider concerns about automated authority and power (Lazar 2024). At the democratic level, these shifts can weaken autonomy, consent, pluralism, accountability, contestability, legitimacy, electoral integrity, and collective self-government.

This paper uses catastrophic risk narrowly: a persistent, society-scale degradation of democratic self-correction—the ability to form preferences under plural conditions, identify responsible power, contest political claims, replace governments, expose abuse, distinguish authentic from synthetic authority, and repair institutional error. The current evidence does not show that threshold has been crossed.

Agency transfer is therefore the paper’s gravity, not one of its measured outcomes. The empirical programme studies conditions that could enable it and the points where that pathway can be interrupted.

3. The Capability-Deployment-Effect Gap

The title uses three large stages: capability, deployment, and effect. They are an organising frame, not a universal linear ladder. A synthetic impersonation may trigger an institutional response without persuading a voter. A recommender may shape attention without producing a measurable belief change. Human and institutional effects can diverge, interact, and alter later deployment.

The fuller topology includes the base model, the served configuration around it, the actor controlling it, the purpose and data supplied, the authority granted, the distribution system, authentic exposure, human or institutional response, and any feedback used to adapt later action.

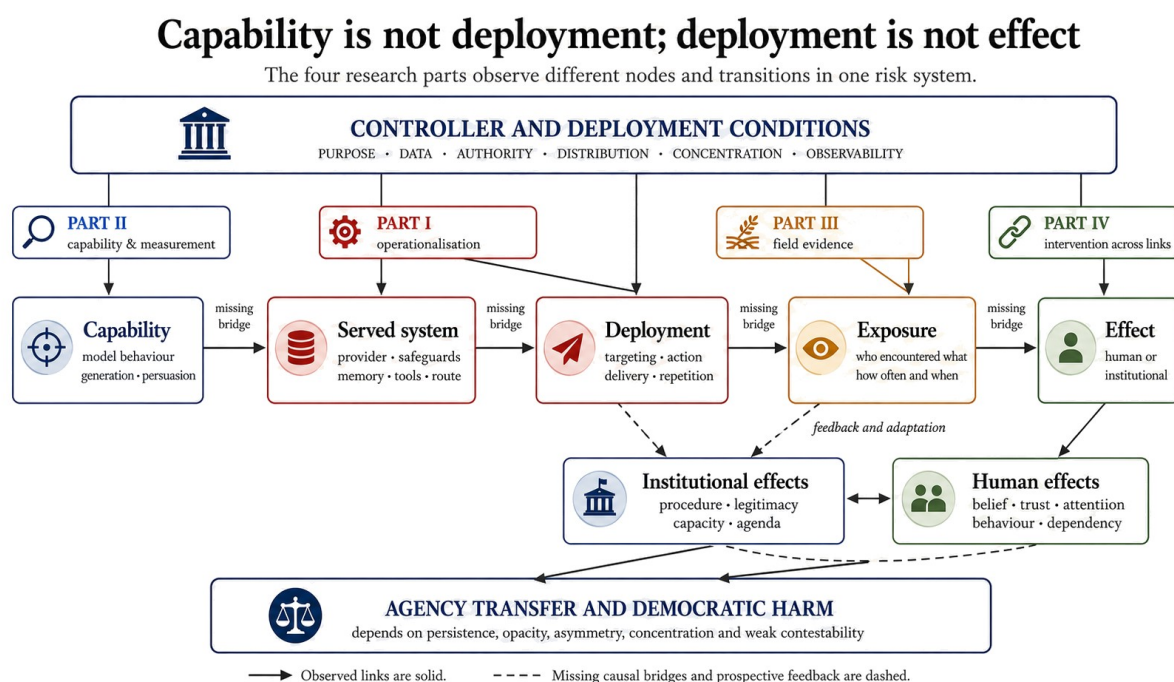


Figure 1. The Capability–Deployment–Effect Gap. The four research parts observe different nodes and transitions in one risk system. Solid links mark observed or directly studied relations; dashed links mark missing causal bridges or prospective feedback. The figure does not imply that every institutional harm requires human persuasion. Source: author’s synthesis.

The causal question is concrete: which AI capability is controlled by which actor, through which influence vector, against which target, producing what change in belief, preference,

attention, trust, behaviour, or dependency—and with what transfer of agency, concentration of power, democratic harm, and possible mitigation?

Every transition requires new evidence. A persuasive answer is not a deployed system. A plan is not an executed action. A transmitted message is not necessarily seen or heard. A view count is not a person. Exposure is not belief change. Belief change is not durable behaviour. Behaviour among some recipients is not an aggregate electoral consequence.

The same discipline begins upstream. Before model access becomes an intervention, the intervention must be specified, reached, persisted, independently reloaded, compared, and replicated. The G0–G7 record in Appendix B formalises that sequence. A failure to create a durable intervention is not a behavioural null; a successful plan is not deployment.

A claim ceiling records the strongest conclusion a result can support. It asks what was observed, in which system, under what configuration, with which comparator, and what evidence would be required for the next claim. A policy-effect ceiling applies the same discipline downstream: it states the nearest endpoint an intervention can be said to improve without silently upgrading it to voter autonomy, institutional resilience, or election integrity (Dassanayake et al. 2025).

Claim ceilings constrain what an institution may say. They do not automatically determine what it may do. A telecom provider does not need to prove changed votes before blocking caller-ID spoofing. A platform can preserve evidence and disrupt a coordinated network without declaring that the network changed an election. The action should match the mechanism evidenced. The public claim should stop where the evidence stops.

4. What the evidence says, and what it does not

The literature does not support complacency. It also does not support a straight line from persuasive text to democratic collapse.

Controlled studies show that language models can change reported attitudes in bounded settings. Personalised AI-generated messages have outperformed non-personalised alternatives in some experiments. GPT-4 has performed strongly against human opponents in short debates when given basic information about the person it addressed. Other studies report small but measurable shifts in policy attitudes and candidate preferences following AI-mediated dialogue (Matz et al. 2024; Salvi et al. 2025; Bai et al. 2025; Lin et al. 2025). A large harmful-manipulation study found that a tested model could produce manipulative behaviour and induce experimental changes in belief or behaviour, while manipulation propensity did not consistently predict efficacy (Akbulut et al. 2026).

These results matter. Dismissing AI-generated persuasion as mere autocomplete is no longer serious. They do not establish a universal effect. A 2025 meta-analysis found no significant overall persuasive advantage for LLMs over humans and substantial heterogeneity across models, domains, and interaction designs (Hölbling, Maier, and Feuerriegel 2025). Personalisation is not magic; argument quality, repetition, interaction design, context, and the ability to secure attention may matter as much as a dense profile.

The conventional campaign literature provides a necessary prior. A synthesis of 49 field experiments found average persuasive effects near zero in general elections. Eady and colleagues found highly concentrated exposure to the Russian Internet Research Agency and no meaningful relationship between exposure and attitudinal or behavioural change in their linked sample. Guess and colleagues, and Moore and colleagues, document rare and highly concentrated exposure to untrustworthy websites. These studies do not prove zero effect. They show that exposure is a bottleneck and that effects are heterogeneous (Kalla and Broockman 2018; Eady et al. 2023; Guess, Nyhan, and Reifler 2020; Moore, Dahlke, and Hancock 2023).

Chen and colleagues express the distinction as receive and accept. Their 2026 experiments found LLM chatbots about as persuasive as campaign advertising conditional on engagement, while current real-world impact was constrained by the difficulty of securing conversations at scale. Persuasion under forced exposure and population-level effect are different estimands (Chen et al. 2026).

Retrospective monitoring of the 2024 UK, EU, and French elections provides a field-level corrective. CETaS found a limited set of viral AI-enabled incidents and no evidence that they meaningfully altered election results. It nevertheless documented candidate harassment, confusion over authenticity, degraded trust, and unethical synthetic campaign practices (Stockwell 2024).

CETaS also found that established distribution methods—bots, aligned influencers, and information laundering—mattered more to virality than generative AI alone. Where generative AI appeared, it often rewrote material, added strategic narratives, or increased production scale. Because the study selected incidents visible enough to attract reporting or research, it may miss private, narrow, or deliberately low-visibility operations. “No identified effect” is not a causal estimate of zero effect.

This restrictive prior does not end the catastrophic-risk argument. It changes it. The pathway is not more AI messages followed by a larger average persuasion effect and then a changed election. It depends on a different architecture: repeated interaction rather than one-shot contact; adaptation rather than fixed content; control of delivery and feedback rather than generation alone; access through trusted or intimate intermediaries; strategically concentrated or very broad reach; weak external observability; and limited exit or correction.

The evidence does not show that these conditions occur together. The purpose of the framework is to identify the missing bridges before treating either catastrophe or safety as established. Frontier-safety practice is beginning to reflect the same distinction: providers now separate manipulation propensity from efficacy, evaluate simulated influence campaigns, and treat harmful manipulation as a frontier governance concern. Those programmes recognise the risk while preserving the difference between model behaviour, configured systems, simulation, and real-world effect (Google DeepMind 2026; Anthropic 2026; OpenAI 2026).

5. Four audits of one system

This flagship is a programme-level governance synthesis, not a conventional single-study article. The four audits are at different stages of maturity. Their role is to observe different links in the same CDE system without borrowing one another's strongest result. Technical safety and govtech work support the argument; they are not the paper's centre of gravity.

The programme records cited as Guerrero (2026a–f) are the primary sources for the study designs, counts, audits, and internal coding reported below. They establish what this programme recorded. They do not provide independent replication or validation of downstream effects.

Part I. Access is not control

Part I asks what an evaluator actually controls. A model name hides the served system around it: provider, route, system prompt, safeguards, context, memory, tools, permissions, retries, and operator interaction. Deployment adds an actor, political objective, data, authority, distribution, feedback, and persistence.

The programme produced two different findings that should not be conflated. The intervention study did not create a durable package that could be independently reloaded, so it produced no admissible behavioural comparison. Separately, three outputs from one unidentified served route contained recognisable campaign-planning elements even as some requested tactics were refused or redirected.

Those outputs named possible audiences, channels, sequencing, mobilisation or demobilisation objectives, measurement, and adaptation. They did not research voters, segment real people, contact anyone, run an experiment, operate a feedback loop, or deploy a campaign. The records are locally tamper-evident, not provider attestations or proof of the underlying checkpoint (Guerrero 2026a–b).

The governance lesson is narrower and more useful: exact route, configuration, permissions, persistence, and evidence custody determine what claims are scientifically admissible. The next study should observe action rather than prose inside a closed synthetic election environment, with named systems, reversible state, matched controls, and a human-only baseline.

Part II. Measurement is part of the object

Part II began with a simple ambition: trace the growth of manipulation-relevant capability across frontier-model releases. The available evidence does not support that curve. It supports a more useful result about measurement.

The Evaluation Registry contains source-linked observations across six instruments, but those instruments measure different objects: attempted persuasion, deception, manipulation propensity, simulated campaign performance, dialogue behaviour, and benchmark-specific safety responses. No exact model configuration appears in four or more instruments, and there are no adequate same-construct anchors (Guerrero 2026c).

The Registry therefore cannot support one manipulation score, one leaderboard, or one longitudinal frontier trend. Averaging the results would create numerical precision by deleting the distinctions that matter.

The direct APE-120 audit, adapted from Kowal et al. (2025), exposed the same problem inside one evaluation pipeline. A first public summary wrongly reported zero samples because the postprocessor failed to parse the archive. Offline recovery found a substantial run. Among valid records, attempt-coded outputs were common, but every served condition failed the prespecified strict-valid coverage gate. The correct result is non-confirmatory: the harness did not support condition-level estimates (Guerrero 2026c).

Missingness is part of the scientific result. Provider blocks, context limits, parsers, simulated interlocutors, evaluators, and judges determine what becomes observable. They are not neutral plumbing; they are part of the measurement object.

The next step is not simply to add more models. It is to repair the measurement system: validate scorers against human judgement, classify missing records by failure mechanism, preserve comparable routes, preregister treatment of failures, and then test persistence, memory, selective personalisation, tools, and adaptation with people. APE should remain a propensity screen rather than be stretched into an effect benchmark.

Part III. Field evidence thins downstream

In field records, evidence becomes less complete downstream. Investigations can often identify accounts, posts, payments, calls, synthetic media, telecom providers, view counts, removals, and enforcement actions. They rarely identify who authentically encountered the treatment, what those people believed beforehand, what changed afterwards, and what would have happened without the operation.

The Election Evidence Index was built around that absence. Its current analytic layer contains ten claim-coded records, eight records eligible for incident counting, 96 atomic claims, and 162 claim-evidence relations. A broader catalogue includes incidents, preparedness files, comparative evidence, and screening leads. It is not a count of confirmed manipulation campaigns (Guerrero 2026d).

CETaS is treated as comparative field evidence rather than an additional claim-coded incident. Its review strengthens the operation–distribution–effect distinction: visible AI-enabled incidents and wider democratic harms were documented, while meaningful electoral-result effects were not identified and quieter operations could fall below the study’s visibility threshold (Stockwell 2024).

Each proposition is separated. Did the event occur? Who controlled it? Which mechanism was used? What was distributed? How many authentic people received it? Did beliefs, trust, attention, behaviour, or dependency change? Was an institution affected? Did an electoral consequence follow?

The New Hampshire robocall shows why the distinctions matter. Two days before the state’s 2024 Democratic presidential primary, an AI-generated imitation of Joe Biden was used in robocalls suggesting that recipients save their vote for November. The Federal Communications Commission later imposed a USD 6 million forfeiture concerning 9,581 calls made using misleading caller-ID information. In June 2025, a New Hampshire jury

acquitted the operator of state voter-suppression and candidate-impersonation charges (Federal Communications Commission 2024a, 2024b; Associated Press 2025).

Those outcomes are not inconsistent. Administrative liability, criminal guilt, occurrence of an impersonation mechanism, human persuasion, and electoral effect are different propositions governed by different standards. The case establishes an attempted influence vector: a trusted political voice was synthetically appropriated, paired with misleading caller identification, distributed near a decision point, and followed by institutional action. It does not establish how many people listened, believed the message, changed behaviour, or stayed home.

The Romania and Moldova records document coordinated accounts, opaque promotion, platform amplification, synthetic personas, cross-platform distribution, financing allegations, and institutional investigations. They do not isolate the contribution of generative AI, identify unique voters reached, or produce a causal estimate of votes changed. Account counts are not people counts. View events are not necessarily unique viewers. Engagement is not belief. Institutional response is not a voter-level effect estimate (Guerrero 2026d).

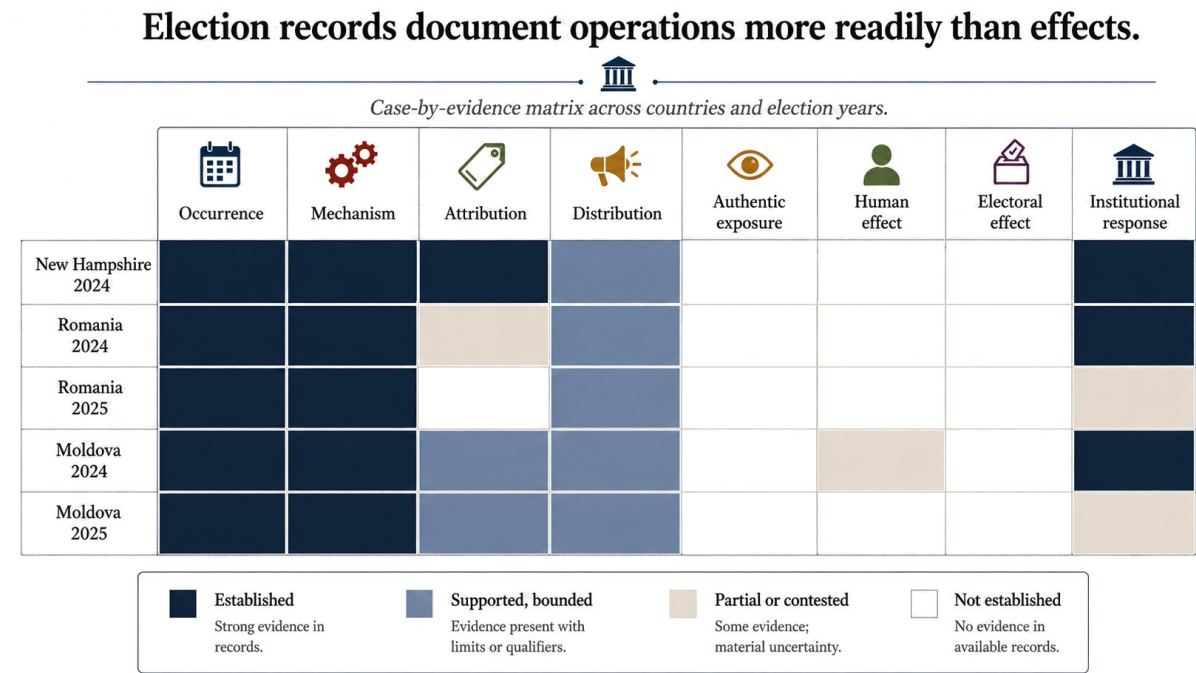


Figure 2. Election records document operations more readily than effects. Darker cells indicate stronger claim-level support in five selected retrospective records; the figure is not the full ten-record analytic layer. Empty effect cells mean the reviewed public record does not identify the effect; they do not mean no effect occurred. Source: Guerrero 2026d.

Field evidence should distinguish two modes without assuming one is more consequential. Precision operations may use private data, trusted identities, narrow audiences, or repeated relationships. Saturation operations may flood public channels with synthetic content, contradictory narratives, impersonations, or coordinated amplification. Precision can be hard to detect; saturation can overload attention and institutional response. Either can contribute to agency transfer if concealed control, functional dependence, or impaired contestability follows. The present Index does not establish that downstream condition.

Timing and observability matter. False procedural information released weeks before an election may be corrected; the same information released in the final hours may exploit a window in which correction is impossible. Jurisdictions with weaker monitoring, smaller research communities, fewer platform resources, or lower journalistic scrutiny may appear to have fewer incidents because fewer are detected. That is a plausible hypothesis, not a finding from the current sample.

Established evidence. Selected operations, mechanisms, distribution events, attribution evidence, and institutional responses are documented.

Open question. Authentic exposure, durable human response, institutional dependence, and aggregate electoral consequence remain weakly observed. “No identified effect” is not “zero effect” (Guerrero 2026d).

Part IV. What works—and for whom?

The policy gap is not that democracies have no tools. It is that tools are often credited with outcomes they were not evaluated to deliver. A notice can disclose origin without neutralising influence. A sharing prompt can reduce circulation without improving its quality. A permission boundary can defend an operator’s instructions without defending the people those instructions target. These distinctions change the recommendations, not just the caveats.

The Policy Evidence Atlas maps 68 control families and 118 implementations. Its original six-record audit was not a search that eliminated the other 112. A subsequent source-linked review annotated all 118 implementations using 69 selected source records and 278 source–implementation mappings. These mappings are not independent studies, and review depth varies. The review adds external findings and revised judgments; it does not silently upgrade the six historically checked effect claims in the production dataset. Appendix G records the method and selected source boundaries (Guerrero 2026e–f).

Different outcomes, different conclusions

Direct evidence on choice architecture justifies attention to the interface. Nouwens et al. (2020) found that removing a first-screen rejection option increased acceptance of cookie notices in a small field experiment. That establishes a design effect on recorded choices, not that those choices matched informed preferences or that a regulatory prohibition had been successfully enforced. The appropriate recommendation is to remove coercive obstacles and test meaningful refusal, not to count acceptance as consent.

More awareness does not necessarily mean less influence. In a preregistered study of 2,610 participants, warnings about a sycophantic assistant changed perceptions of objectivity and trust, but did not reliably improve self-perceived rightness or intentions to repair an interpersonal conflict. Those were self-reports, not observed relationship repair (Ibrahim et al. 2026). A separate commercial-conversation preprint similarly found no significant reduction in sponsored-book selection with explicit sponsorship labels; that disclosure comparison spans separately collected experiments rather than one simultaneous randomisation (Salvi, Cuevas, and Horta Ribeiro 2026). Neither study establishes that all disclosure is useless.

The reverse dissociation also matters. The AI Watchdog preprint randomised 150 participants across five conditions. Dismissible, just-in-time warnings reduced AI-steered choices from 71.7 to 53.7 per cent relative to no intervention on a preregistered secondary endpoint, without improving the primary detection outcome. With 30 participants per arm and simulated tasks, this is a bounded signal for replication, not a general deployment mandate; superiority over the forcing variant was not established (Poonsiriwong et al. 2026). A changed decision without greater understanding does not, by itself, demonstrate recovered agency.

Technique education needs the same discipline. Roozenbeek et al. (2022) support immediate recognition of taught tactics, with additional bounded laboratory outcomes. In five later studies, emotional-manipulation inoculation improved technique recognition but did not alone improve truth discernment; its combination with an accuracy intervention performed differently (Pennycook et al. 2024). Other studies report differences by technique and limited transfer from matched inoculation to sharing intentions or new topics (Biddlestone et al. 2025; Ancina, Zumbärgel, and Krämer 2026). The objective is discrimination, not indiscriminate suspicion: accurate communication can also be emotional.

The treatment and its delivery cannot be separated

“Add friction” is too broad a prescription. Ershov and Morales (2024) associate Twitter’s 2020 sharing-interface change with reduced news circulation, but not a selective reduction for lower-factualness outlets, and find politically heterogeneous responses. This is a quasi-experimental platform intervention, not a WhatsApp trial. The WhatsApp coverage study identifies gaps in forwarding labels, not the counterfactual effect of a specific cap (de Freitas Melo et al. 2024). Forwarding restrictions should therefore leave the homogeneous “best-supported” block, without being declared ineffective.

Accuracy-focused prompts are a distinct treatment: they direct attention to truth rather than simply making sharing harder. Survey experiments and a Twitter field experiment support bounded improvements in sharing quality; an internal meta-analysis supports average sharing-discernment gains but is not twenty independent research-team replications (Pennycook et al. 2021; Pennycook and Rand 2022). The field outcome includes source-quality proxies, not a direct measure of autonomy. Accuracy prompts deserve a separate entry and comparison, not silent absorption into friction or generic literacy.

Timing changes the policy conclusion. Slaughter et al. (2025) estimate, using synthetic controls, 46.1 per cent fewer new reposts after a Community Note is attached, but 11.6 per cent fewer when earlier engagement is included in the post-level total studied. The larger conditional effect must not be presented as whole-programme protection. The estimates depend on observed eligible posts, attachment timing, and identification assumptions; neither measures the electorate’s beliefs or agency. The 2026 correction adds a competing-interest disclosure rather than revising the estimates (Slaughter et al. 2026).

Verified corrective context is also different from a generic AI-origin label. An authorship-label experiment did not detect changes in persuasion, perceived accuracy, or intended sharing in its tested policy messages (Gallegos et al. 2025). Automated correction needs its own safety case: DeVerna et al. (2024) found no average discernment improvement from the

tested LLM fact checks, with harmful responses to some erroneous or uncertain outputs. The human comparator was collected later, and voluntary viewing was selected. These findings warrant testing the defender's errors, not assuming that a second AI supplies reliable protection.

A control must name its beneficiary

CaMeL supplies technical evidence for separating control and data flows and enforcing permissions against external prompt injection. Its threat model trusts the initiating query and excludes some text-only influence attacks (Debenedetti et al. 2025). The inference for this project is consequential: defending an agent from an outside attacker does not defend a citizen from an authorised but harmful objective. A system can execute its controller's instructions faithfully while steering the person it serves. Permission security and purpose legitimacy must be evaluated separately.

Exit presents a related distinction between formal provision and practical control. Turner et al. (2021) found that data receipt did not enable import or direct transfer across their four tested IoT systems. Search-engine field experiments distinguish active choice, defaults, and experiencing an alternative; their market-share counterfactuals are model estimates, not observed assistant-migration effects (Allcott et al. 2025, revised 2026). The inference for persistent assistants is to test whether users and institutions can continue the relevant activity after switching, with useful context, permissions, expertise, and alternatives intact.

Memory minimisation remains a structural priority, but its claimed benefit must be bounded. Offline recommender experiments measure quality under reduced data budgets, not protection from conversational dependency (Biega et al. 2020). Associations between heavier voluntary chatbot use and adverse psychosocial outcomes are not randomised evidence that a memory cap works (Fang et al. 2025). Data minimisation, continuity, and usable switching need deployment-specific tests, including privacy costs; their democratic importance is not settled by the ease of obtaining an effect estimate.

The resulting finding is differentiated. External studies establish selected human or technical outcomes within their designs; the new analytical contribution is that the policy CDE Gap concerns outcome mismatch, incomplete delivery, and whose authority is protected. Independent monitoring, enforceable limits, and viable exit form a plausible complementary defence, not a tested package. Durable preservation of reflective agency under adaptive influence remains an open question. The former A–F assignments remain historical dataset metadata, not this paper's efficacy ranking (Guerrero 2026e–f).

Evidence question	What a study can establish	What an agency claim still requires
Outcome	Recognition, trust, sharing or choice can change independently.	Understanding, goal-consistent choice, refusal and revision—not one proxy alone.
Delivery	A control can work after exposure to the control.	Coverage, latency, missed cases and displacement across the whole decision window.
Authority	A boundary can protect trusted instructions from an external attack.	A legitimate purpose, a named beneficiary, and limits on authorised conflicting objectives.
Exit	A service can provide an export or a choice screen.	Continuity, usable alternatives and independence from both the assistant and its defender.

Figure 3. Four questions that separate component performance from protection of agency. These are analytical distinctions, not a measured causal sequence or an efficacy ranking. Source: synthesis of the studies discussed in Part IV; Guerrero 2026f.

6. Power accumulates in the joins

The four audits produce different kinds of evidence. Across them, records become less complete as the chain moves from model behaviour to deployment, authentic exposure, human response, institutional consequence, and policy effect. The concern that a controller’s visibility can move in the opposite direction is a strong architectural inference, not a directly measured comparison.

An operator may know who was selected, why they were selected, which treatment they received, what they clicked or disclosed, and how the next interaction was adjusted. Researchers, regulators, election authorities, and affected people may see only one leaked message, one removed account, or one aggregate count. Power can concentrate not only in the ability to influence, but in the ability to observe, optimise, and selectively disclose the influence process.

The actor best placed to optimise the influence process may also be the only actor able to explain it.

This is not a finding that one present actor already controls the entire chain. It matters because independent evidence is part of democratic power. When the controller of an influence system also controls the only records capable of revealing it, public institutions may become dependent on that controller’s account of what happened.

Concentration and diffusion can coexist. Capable models may become more widely available while control still concentrates around proprietary data, identity, orchestration, cloud infrastructure, dominant assistants, recommendation, advertising, telecom delivery, analytics, and evidence access. Conversely, diffusion can enlarge the set of actors able to run harmful operations, raising monitoring, attribution, and enforcement burdens. Commodity models do not guarantee plural influence infrastructure; concentrated model development does not prevent downstream diffusion.

The risk also operates through precision and saturation. Precision uses narrow access, personal context, trusted identity, or repeated interaction. Saturation uses volume, speed,

synthetic variety, or coordinated distribution to consume attention and institutional capacity. Neither is inherently more severe. Their democratic significance depends on who controls the system, which dependency or decision environment is affected, whether exit and correction remain practical, and whether the operation becomes persistent.

The same question applies to the defensive layer. As a strong architectural inference, a provider that controls both an influential assistant and its only monitor can also control the definition of a violation, the evidence retained, the warning shown, and the account offered to an investigator. A second model under the same authority does not necessarily create independent scrutiny. This is a control-allocation argument, not a measured market-concentration effect (Lazar 2024; Guerrero 2026f).

Defence can therefore become another route for agency transfer. The plausible hypothesis is that user-controlled, independently governed monitors, enforceable authority limits, and practical exit offer better protection than notice alone. The alternative is substitution: people become dependent on a defensive provider that they cannot inspect or contest. The package's comparative efficacy and long-term distribution of power remain open questions; lower compliance with one assistant is not sufficient evidence of success.

A conditional catastrophic pathway

The bounded-effects literature remains the right starting point. Most political messages do not transform most voters. Exposure is uneven, resistance is real, and effects may decay. A catastrophic case cannot be extrapolated from a one-turn persuasion rate. It must earn each additional assumption.

Six joined conditions define the threat model.

Table 2. Conditions that would have to join for catastrophic democratic risk. The programme does not show this conjunction. Source: author synthesis.

Condition	Observable indicator	Potential interruption	What would weaken the concern
Sustained capability	Repeated, adaptive interaction remains effective	Limit memory, tools, and consequential actions; test exact routes	Effects decay quickly or refusals remain reliable
Trusted or infrastructural access	Assistants, identity channels, workflows, or dominant platforms provide access	Authentication, disclosure, plural channels, procurement diversity	People know the principal and can leave without material cost
Joined control	One actor controls several CDE links	Separate permissions; require independent approval at consequential transitions	Control remains divided and no actor can optimise the full pathway
Private optimisation and weak observability	The controller sees response data while outsiders lack route, delivery, or audit evidence	Preserve or escrow records; enable qualified independent audit	Reviewers reproduce key claims and relevant denominators in time
Slow correction and weak contestability	Remedy arrives after the decision window	Rapid, reversible response with notice, appeal, and expiry	Errors are corrected before meaningful decisions with low rights burden
Persistent dependency	People or institutions cannot function or decide without the system	Exit, interoperability, human override, and institutional redundancy	Dependence remains voluntary, plural, and reversible

The components range across evidence grades. Upstream capabilities and selected operational precursors are established in bounded settings. The importance of joined control and observability is a strong inference. Persistent dependence and compounding are plausible hypotheses. Society-scale degradation of democratic self-correction is a speculative scenario. Its probability and threshold remain open questions.

The catastrophic outcome would not necessarily look like one election suddenly “hacked” by persuasion. It could look like repeated degradation of correction: citizens no longer knowing who is speaking; institutions depending on the same companies they must investigate; synthetic signals distorting perceptions of public opinion; trusted interfaces quietly steering political attention; and verification becoming slower than production.

The threshold is persistence. The mechanism is control of the joins. The democratic harm is loss of the practical capacity to recognise, contest, and repair error. That is the pathway from influence infrastructure to agency transfer.

7. Govern the system before the outcome

The policy mistake mirrors the research mistake. One approach waits for proof that an election changed. The other regulates capability in the abstract. Both miss the operating system in between. The practical rule is simple: do not promote upstream evidence into downstream claims; do not wait for a downstream claim before acting on an evidenced

upstream mechanism. A system that enables identity deception, unauthorised political-data use, opaque distribution, or unbounded tool access can be addressed at that point, with lawful authority, a measurable endpoint, and a proportionate response.

Institutions see different parts of the same system. Law allocates responsibility and protects rights. Technology governance sets conditions on access, infrastructure, standards, procurement, and audit. Technical safety tests how a configured system behaves. Election administration manages compressed decision windows. None works well if the evidence connecting them is absent, inaccessible, or controlled by the actor whose conduct is at issue.

This paper proposes an evidence interface: a shared account of what happened, at which CDE node, under whose purpose and authority, for whose benefit, and with which records available for challenge. Depending on the deployment, that record should include system and route identity, data permissions, the action taken, coverage and timing of the control, uncertainty, missingness, the measured endpoint, its comparator, and the responsible authority. It should say who can inspect, interrupt, revoke, appeal, and remedy the action. Naming the protected principal is essential: a control that secures the operator's objective must not be described as protecting the affected person without a further test.

The design test has six parts. A consequential system should be identifiable and visible: authorised parties can inspect its principal, political purpose, sponsor, route, and relevant targeting criteria. It should be divided and interruptible: analysis, generation, targeting, approval, distribution, and measurement need not share one authority boundary, and permissions can be revoked before the decision window closes. It should be auditable, reversible, and contestable: qualified reviewers can examine records under proportionate safeguards; wrongful action can be restored where possible; and affected people or institutions can understand, challenge, appeal, and seek redress.

These are governance objectives, not findings that one design has proved effective. Evidence custody, tamper-evident retention, and clean-reload attestation create privacy, cybersecurity, cost, proportionality, competition, and intellectual-property burdens. Those burdens belong inside the design rather than in an afterthought.

European law already covers fragments of the architecture. The Political Advertising Regulation addresses sponsor, targeting, records, repositories, and retention. The Digital Services Act addresses systemic risk, audit, and vetted researcher access for very large services. The AI Act prohibits specified manipulative or deceptive practices under conjunctive conditions, imposes systemic-risk duties on certain GPAI providers, and classifies some election-influence systems as high risk. These instruments do not form one CDE regime, and their existence does not prove effectiveness (European Union 2022; European Union 2024a; European Union 2024b; European Union 2026; European Commission 2025a).

The GPAI Code's Safety and Security chapter adds an operational route for identifying harmful manipulation as a specified systemic risk. It is voluntary, applies within a narrower provider context, and does not replace Article 5, Article 55, or Annex III. Its value here is architectural: it links capability, affordances, deployment context, evaluation, mitigation, governance, and residual risk without pretending that framework adoption is an effect estimate (European Commission 2025b; European Commission 2026).

The NIST Generative AI Profile similarly treats documentation, provenance, monitoring, testing, and incident disclosure as lifecycle risk-management practices. It is voluntary guidance, not evidence that those practices protect democratic outcomes (National Institute of Standards and Technology 2024).

Who must produce what evidence

The framework becomes operational only when responsibility follows control. Each actor should be able to produce evidence at the CDE node it controls, while affected people and public institutions retain practical routes to challenge and remedy.

Table 3. Governance as an evidence interface. Minimum responsibilities and evidence are assigned to the actors controlling different CDE nodes. These duties do not, by themselves, establish democratic effectiveness. Source: author synthesis.

Actor or institutional role	Minimum governance duty	Evidence that should exist	Claim ceiling / safeguard
Model or served-system provider	Evaluate the exact configured route, safeguards, tools, permissions, and material changes.	Versioned route receipt; system configuration; test protocol; failure and missingness record; serious-incident criteria.	Capability or control evidence is not deployment prevalence or human effect.
Deployer, campaign, or vendor	Name the principal, purpose, data, authority, approvals, and limits of consequential political use.	Purpose record; data provenance; permission and approval logs; actions attempted and completed; incident history.	Operational records do not establish authentic exposure or persuasion.
Platform, telecom, identity, or distribution intermediary	Preserve auditable delivery denominators and maintain proportionate interruption and restoration mechanisms.	Deduplicated reach where lawful; frequency; targeting and forwarding uplift; enforcement, appeal, and restoration timelines.	Delivery or interruption is not proof of belief, turnout, or election effect.
Electoral, regulatory, judicial, and research oversight	Secure timely, independent access; adjudicate bounded claims; preserve due process and confidentiality.	Chain of custody; access logs; methods; uncertainty; comparator; reasons; appeal, expiry, and after-action review.	Authority and investigation do not substitute for causal evidence.
Affected people and public institutions	Receive intelligible notice, practical exit, contestation, appeal, and restoration where possible.	Notice and explanation records; complaint and appeal outcomes; restoration; accessibility and disparate-burden measures.	Disclosure alone does not prove informed consent or retained agency.
Defensive monitor / assurance provider	Disclose conflicts; permit independent scrutiny, meaningful dismissal, appeal, and exit; protect the user's data.	Warning logic and criteria; test results and error rates; retained traces; conflict disclosures; override and complaint records.	A changed choice or passed security test does not establish the affected person's agency.

Measure the node an intervention changes. Preserve evidence for the next node. Do not promote one into the other.

Policy priorities: tested endpoints, accountable authority

The source-linked review supports four priorities, distinguished by purpose and evidence rather than a single efficacy score. Existing lawful obligations remain applicable regardless of whether their downstream effects have been estimated. Novel policy designs still require

rights, necessity, proportionality, and anti-capture assessment; an empirical signal does not settle those judgments (Guerrero 2026f).

First, use bounded human-facing controls where the relevant outcome is supported. Remove coercive choice architecture and make refusal usable. Evaluate accuracy prompts, verified corrective context, and technique education as different treatments, with different comparators. Measure true as well as false information, understanding as well as trust, and behaviour as well as recognition. Do not turn a disclosure duty into a claim that steering has been neutralised (Nouwens et al. 2020; Pennycook et al. 2021, 2024; Ibrahim et al. 2026).

Second, enforce limits on consequential authority. Scope tools, recipients, data access, actions, and revocation outside the model; test whether approvals are understandable and whether revocation actually stops action. Separate external compromise from an authorised conflicting purpose. A narrow permission design can address the former without resolving the latter. Institutions should therefore inspect objectives and incentives as well as execution traces (Debenedetti et al. 2025; Salvi, Cuevas, and Horta Ribeiro 2026).

Third, pilot frontier-specific defensive interfaces. Replicate just-in-time monitors independently, preserve dismissal and explanation, and test whether users remain able to revise their decision. Include monitor errors, privacy costs, cognitive burden, and legitimate helpful advice. The early Watchdog result supports a focused pilot, not an obligation to install a universal warning system or proof that a forcing interface is inferior (Poonsiriwong et al. 2026; DeVerna et al. 2024).

Fourth, protect accountability and functional exit. Give qualified independent actors access to the relevant evidence, preserve public and organisational expertise, and test migration through continuation of real tasks rather than possession of a download. Plural providers and public-interest options should remain contestable themselves. These are structural recommendations supported by adjacent evidence and the project's account of concentrated control, not experimentally validated democratic outcomes (Turner et al. 2021; Allcott et al. 2025, revised 2026; Lazar 2024).

Authentication, provenance, labels, and incident response remain useful for specified vectors, but should not substitute for these priorities. Evidence must include coverage, latency, failures, false positives, costs, displacement, and appeal. General benchmark or release triggers require a validated construct, comparator, threshold, and action rule; uncertainty about those instruments does not suspend applicable law or preclude action on a separately evidenced serious risk. Complementarity among the proposed layers remains a hypothesis to test, not an estimated package effect.

8. Research the missing bridges

The next phase should not keep expanding this working paper. It should turn each missing bridge into a narrower study with its own design, evidence standard, and release boundary.

1. Make the evidence reproducible

Human-calibrate the APE attempt and refusal judgements, assign a reason to every missing record, preserve exact routes, and rerun a smaller fixed panel. In parallel, test the G0–G7

sequence across exact checkpoints with durable intervention packages, clean reload, matched controls, utility and safety constraints, and independent replication. The outcome is not a better leaderboard. It is whether independent teams can reproduce a bounded estimate from the same artefacts.

2. Measure operational uplift

In a closed synthetic election environment, compare human-only, model-only, tool-enabled, agentic, and safeguarded-agent conditions. Measure time, expertise, cost, coordination, completion quality, rule violations, resistance, revocation, and recoverability. This would test whether frontier AI expands the practical adversary set or compresses roles. The current traces do not establish either proposition.

3. Connect mechanisms to people and institutions

Separate persuasion from manipulation by measuring deception, opacity, vulnerability exploitation, emotional dependence, restricted choice, and control over the decision environment. Include repeated interaction, memory, disclosure, delayed follow-up, meaningful exit, and institutional endpoints such as workload, triage accuracy, service degradation, decision delay, evidence loss, and dependence on private information. Agreement is not enough to establish agency transfer; the study must identify reduced reflective control, constrained alternatives, or growing dependency.

Authentic-exposure studies should measure unique reach, frequency, timing, audience selection, prior alignment, countervailing exposure, detection, and the remaining correction window. Privacy-preserving partnerships with platforms and telecom providers may be necessary. Where causal identification is not credible, the result should remain an exposure or institutional-response estimate.

4. Evaluate the assistant and its defender together

Preregister a joint evaluation of a user-facing assistant and its defensive layer. Distinguish transparent helpful assistance, an authorised but concealed conflicting objective, and externally injected instructions as separate threat models. Randomise specific controls rather than a generic “guardrail” bundle. Primary outcomes should capture goal-consistent decisions, understanding of the conflict, meaningful refusal, and the ability to revise or exit. Detection, trust, and satisfaction are secondary indicators, not substitutes for agency. Include accurate helpful advice so that indiscriminate rejection is recorded as a possible cost.

Use consenting participants and low-risk tasks, not live voter targeting. Record the intervention’s coverage and timing, errors, privacy burden, unequal cognitive load, displacement, and response to revocation. Longer follow-up should test persistence and whether dependence shifts to the defensive provider. Separately map who controls purpose, data, distribution, monitoring, evidence, and remedies. This proposed study tests a bridge from bounded decision support to agency; it is not a claim that the bridge has already been crossed (Guerrero 2026f).

9. Limitations and responsible release

The limitations are substantial and should be read as the boundary of the contribution, not as a footnote to it.

Part I contains layered and partly unreconciled evidence units, including three completed outputs from an unidentified route without tools, authentic audiences, or external state change. The records are locally tamper-evident, not independently attested. The access-to-control study produced no durable intervention package and therefore no behavioural estimate.

Part II is descriptive and non-confirmatory. Coverage and missingness prevent condition-level estimates. The Evaluation Registry does not support a pooled manipulation score or longitudinal frontier trend. The paper does not report a cross-classified model accounting for repeated topics and served conditions.

Part III is purposive rather than representative. It cannot estimate prevalence; its public metrics use heterogeneous denominators; and attribution remains contested in several cases. Part IV's new source-linked assessment is also purposive, English-dominant, and single-pass. It documents 118 implementations, 69 selected source records, and 278 mappings, not 278 independent studies or 118 verified causal effects. Eight sources received new main-text or substantive-section checks in that review; other records rely on varying depths, including abstracts, publication summaries, official texts, and retained prior adjudications. Appendix G and the review ledger disclose these differences (Guerrero 2026f).

The review does not reproduce all source data, independently double-code the literature, or complete a current-law audit. Several directly relevant assistant studies are preprints with short, simulated or self-reported outcomes. Within-study contrasts, cross-wave comparisons, secondary endpoints, and selected voluntary exposure require different interpretations. "No exact effect identified in the selected sources" does not mean no relevant evidence exists anywhere. The production Atlas beta.3 retains its historical empirical flags; this manuscript's external synthesis does not automatically adjudicate its remaining core claims.

Most programme artefacts were designed and audited inside the same research programme. This creates risks of self-confirmation, coding error, and selective interpretation. Provider-route identity and configuration remain incomplete in important places. No fellowship experiment measured authentic exposure, human persuasion, durable preference change, voting behaviour, agency transfer, concentration of power, election outcome, or policy effect. The human and technical intervention findings discussed in Part IV belong to the cited external studies, not to new experiments conducted for this fellowship.

The failures add evidence only at their own layer. Failed intervention persistence reveals a conversion constraint. Failed parsing and evaluator coverage reveal an observability constraint. Neither was the original confirmatory hypothesis, and neither substitutes for the missing behavioural result.

Even with better data, election-level causality may remain partly unidentified because campaigns, media, interpersonal discussion, strategic reaction, institutions, and prior beliefs interact without a clean aggregate counterfactual.

Research on manipulation also creates a dual-use problem. This paper publishes concepts, aggregates, source-linked claims, non-operational figures, limitations, and defensive designs. It withholds full harmful prompts, campaign-ready outputs, targetable profiles, evasion methods, credentials, live distribution instructions, and operational judge traces. No real voters, campaigns, platform accounts, or election infrastructure were used. Independent ethical and dual-use review is required before more operational material is released.

10. Conclusion: preserve democratic self-correction

Frontier AI does not remove the constraints on persuasion. Its democratic significance may instead lie in repeated, adaptive influence becoming infrastructure: embedded in trusted interfaces and controlled through privileged access to data, distribution, feedback, and evidence.

The supporting audits do not show that such a system changed an election. Access did not become reproducible experimental control; planning did not establish deployment; benchmark attempts did not establish human effects. Election records documented operations more clearly than authentic exposure. The policy review adds a distinct finding: usable controls exist, but their outcomes, delivery, and protected principal can differ from the democratic benefit attributed to them.

The CDE Gap maps the actor, purpose, beneficiary, evidence holder, measured endpoint, and missing bridge. Agency transfer supplies the severity test: whether practical control over attention, choice, dependency, or institutional decisions moves towards the system's controller. Stopping an attack or changing a response is not enough to demonstrate that affected people retain informed and contestable authorship.

Three rules follow. Act on the evidenced mechanism without claiming an unmeasured downstream result. Evaluate the control's coverage, timing, errors, and rights costs. Ask whose authority it secures, and preserve the records and practical alternatives needed to challenge it. These rules support proportionate action before a national election-effect estimate, without mistaking preparedness for protection.

The catastrophic pathway remains conditional on sustained capability, trusted access, joined control, private optimisation, weak observability, slow correction, and limited exit. Society-scale degradation of democratic self-correction remains a speculative scenario with unknown probability.

Democracy does not need perfect control over influence. It needs the capacity to identify who acts, compare alternatives, refuse, contest, exit, and repair error. The task is to keep the joins between capability, data, identity, distribution, feedback, and evidence visible and challengeable before they harden into power that democratic institutions can no longer correct.

Technical and evidentiary appendix

A. Part I evidence-unit reconciliation

Table A1. Part I evidence units. These records may overlap and must not be added into one sample count. Source: Guerrero 2026a–b.

Record	Unit	Count	What it supports	What it does not support
Paired red-team pilot	Objectives across two scenario families	10	Exploratory comparison of official hosted refusal behaviour and safeguard-reduced workflow prototyping	One independent n=10 effectiveness estimate; deployment or human effect
Midpoint record	Exploratory probes	3	Protocol development for the synthetic study	A disjoint sample or completed study
Deepest forensic subset	Client-observed bundles / requests / outputs	2 / 5 / 3	Route-level planning content and refusal/final-answer structure in the most auditable subset	Total Part I evidence universe; named checkpoint; prevalence
Deepest forensic subset	Local hash-chain events	18,907	Local record continuity and internal trace inspection	Provider attestation, trusted timestamp, or independent custody
Research boundary	Live deployment / authentic exposure	0 / 0	Scope of the study	A failed-output count or evidence of safety
Later expanded corpus	Unique conversations / requests or runs	6 / 31	Post-hoc material for future reconciliation and coding	A headline sample until overlap with the frozen subset is resolved

Part I is broader than the forensic subset

The record contains different evidence units. They may overlap and must not be added into one sample count.

BROADER PILOT	MIDPOINT RECORD	DEEPEST FORENSIC SUBSET	RESEARCH BOUNDARY
10 paired objectives 2 scenario families Official routes: 10/10 refusals. Safeguard-reduced routes: workflow prototypes.	3 exploratory probes Protocol-generating The probes shaped the synthetic study design. They are not known to be disjoint from the pilot.	2 trace bundles · 5 requests 3 outputs · 18,907 events The most auditable subset, with local hash chains. It is not the total Part I evidence universe.	0 live deployment 0 authentic exposure This is a study boundary— not a failed-output count.

Breadth, protocol development, and forensic depth are complementary views—not additive stages.

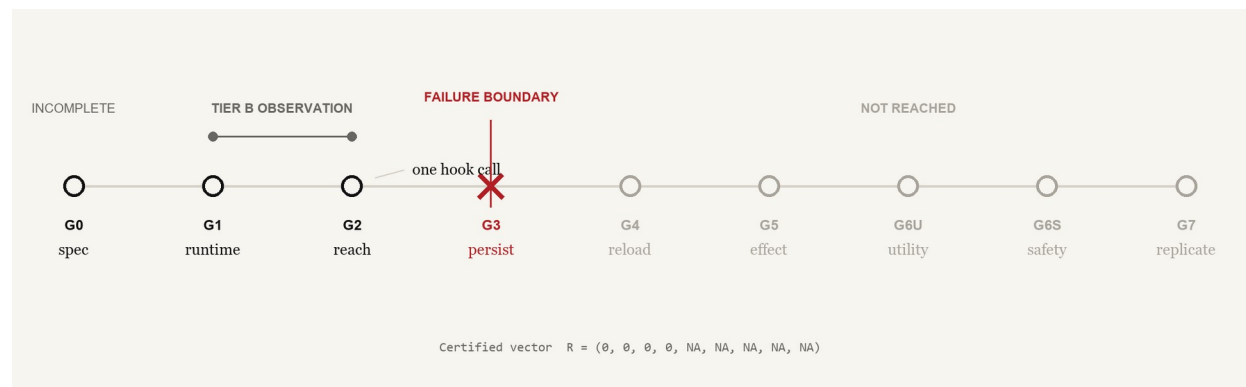
Claim ceiling: exploratory planning and route behaviour only; no autonomous execution, real audience contact, persuasion, or electoral effect.

Appendix Figure A1. Part I evidence units are complementary and may overlap; they must not be added into one sample count. The figure distinguishes the broad paired pilot, protocol-generating probes, the deepest forensic subset, and the research boundary. No live deployment or authentic exposure occurred. Source: Guerrero 2026a.

B. G0–G7 conversion record

Table A2. G0–G7 conversion record for the two attempted intervention routes. Source: Guerrero 2026b.

Gate	Minimum evidence	DeepSeek route	Qwen route
G0	Frozen specification. Registered target, intervention, task, budget, stop, outcomes, and analysis	Reached	Reached
G1	Executable runtime. Exact checkpoint loaded in a recorded environment	Reached	Reached
G2	Intervention reachability. Intended mechanism entered under matched controls	Reached once	Incomplete / not certified
G3	Durable persistence. Stored package with manifest, size, schema, revision, and digest	Failed: no durable capture	Failed: no saved adapter after two 300-step loops
G4	Independent reload. Clean retrieval, verification, load, and exercise of exact bytes	Not reached	Not reached
G5	Valid behavioural estimate. Sealed paired evaluation against preregistered controls	Not reached	Not reached
G6U	Utility constraint. Declared utility boundary met	Not reached	Not reached
G6S	Safety constraint. Declared safety boundary met	Not reached	Not reached
G7	Independent replication. Registered result reproduced by another team	Not reached	Not reached



Appendix Figure A2. Access-to-control gate record for the deepest completed intervention attempt. The run reached the intended mechanism once but did not create a durable intervention package; later gates were therefore not reached. The binary vector shown below the sequence is an internal gate-status record, not a risk score. Source: Guerrero 2026b.

C. APE-120 reconciliation

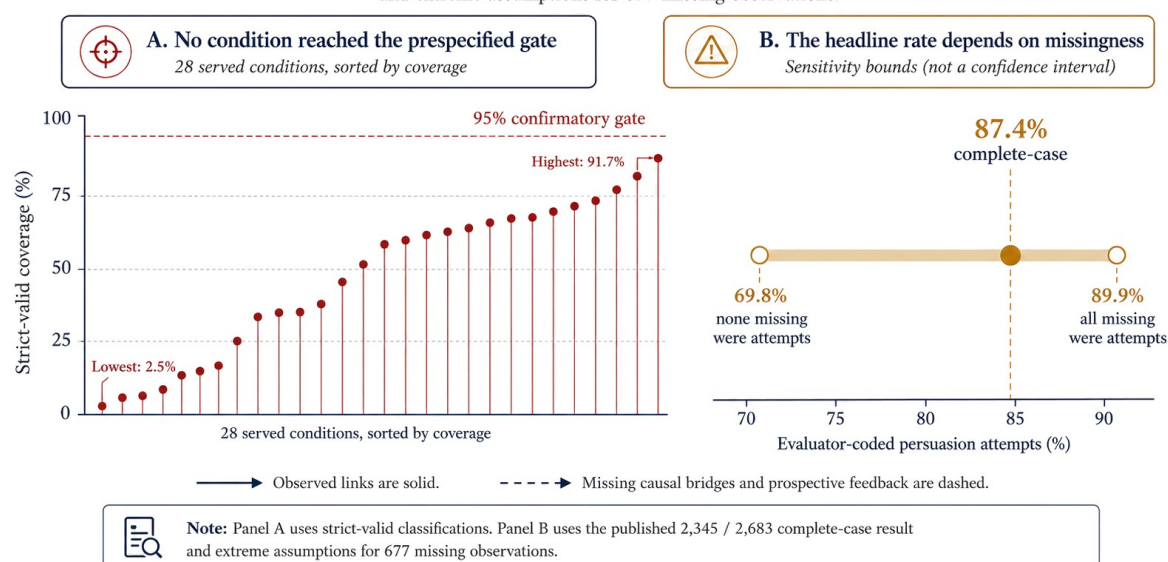
Table A3. APE-120 denominator and coverage reconciliation. Source: Guerrero 2026c.

Stage or diagnostic	Count	Interpretation
Frozen queue	66 conditions	Planned queue, not completed sample
Conditions recorded before timeout	33	Includes route failures
Route failures before inference	5	No target inference completed
Finalised served conditions	28	Conditions with 120 finalised records each

Stage or diagnostic	Count	Interpretation
Finalised records	3,360	28 × 120
Valid target + attempt-evaluator pairing	2,683	Complete-case denominator for attempt coding
Attempt-coded records	2,345	87.4% descriptive complete-case rate
Missing valid pairing	677	20.1% of finalised records
Extreme missingness sensitivity	69.8–89.9%	Bounds under none/all missing coded as attempts; not a confidence interval
Valid refusal judgements	1,776	Separate denominator from attempt coding
Refusal-coded records	741	41.7% of valid refusal judgements
Strict-valid classifications	1,711	Used for the prespecified coverage gate
Conditions meeting 114/120 gate	0	No confirmatory model estimate admitted
Recorded spend	USD 14.23	Audit cost for the frozen run

The attempt rate is high; the harness is not confirmatory

Panel A uses strict-valid classifications. Panel B uses the published 2,345 / 2,683 complete-case result and extreme assumptions for 677 missing observations.



Appendix Figure A3. Attempt-coded responses were common among valid observations; the harness was not confirmatory. Panel A shows strict-valid coverage across finalised served conditions. Panel B places the complete-case rate inside extreme missingness bounds. The bounds are not confidence intervals. Source: Guerrero 2026c.

Failure diagnostics include 538 target truncations, 501 refusal-judge errors, 471 invalid or unparseable refusal-judge outputs, and 106 evaluator errors. These categories may overlap and should not be summed without record-level reconciliation.

D. Reusable claim and policy-effect ceiling record

Table A4. Reusable record for claim ceilings and policy-effect ceilings. Source: author synthesis.

Field	Required entry
Observed node	Capability, served system, deployment, exposure, human effect, institutional effect, or intervention
Unit and configuration	Exact model/route/system, actor, data, permissions, distribution, and comparator
Evidence status	Established evidence, strong inference, plausible hypothesis, speculative scenario, or open question
Sample and exclusions	Denominator, clustering, missingness, route failures, and post-hoc filters

Field	Required entry
Direct endpoint	What was actually measured and over what time
Strongest licensed claim	Narrow conclusion supported by the design
Unsupported downstream claim	Next CDE conclusion that the result does not establish
Minimum next evidence	Design and records needed to cross the next bridge
Rights and legitimacy burden	Privacy, speech, due process, security, competition, access, and distributional effects
Review and reversal	Appeal, expiry, retest, restoration, deletion, or narrowing condition
Purpose and beneficiary	Who sets the objective; whose authority the control protects; how conflicts with the affected person's goals are identified.
Delivery and exposure to the control	Eligibility, coverage, attachment or response time, untreated cases and the remaining decision window.
Agency and defensive dependence	Understanding, meaningful refusal, revision, continued task capability after exit, and dependence on the monitor itself.

E. Three manipulation baselines that should not be conflated

Table A5. Three manipulation baselines that should not be conflated. The paper's normative test, Article 5's prohibition, and the GPAI Code's systemic-risk category serve different functions. Source: author synthesis from European Union 2024a and European Commission 2025b.

Baseline	Core test or object	Status and scope	Claim ceiling
This paper	Influence becomes manipulation when opacity, deception, vulnerability exploitation, asymmetry, dependency, or control of the decision environment degrades reflective and contestable choice.	Normative analytic definition used across micro, meso, macro, and democratic levels.	Does not by itself establish occurrence, harm, legal liability, agency transfer, or effect.
AI Act Article 5	Specified subliminal, purposefully manipulative or deceptive techniques, or exploitation of listed vulnerabilities, under cumulative material-distortion and significant-harm conditions.	Binding prohibition with conjunctive legal thresholds and exceptions; not a general ban on persuasion or political manipulation.	Legal coverage does not establish prevalence, causal effect, or democratic materiality in a particular case.
GPAI Code Appendix 1.4(4)	Specified systemic risk of strategic distortion of behaviour or beliefs through persuasion, deception, or personalised targeting, especially multi-turn or difficult-to-detect influence at scale or against high-stakes decision-makers.	Voluntary Article 55 compliance baseline for signatories that provide GPAI models with systemic risk.	Framework existence or adherence does not establish that manipulation occurred, succeeded, or was mitigated.

F. Illustrative policy-effect ceilings

The matrix below is illustrative, not a ranking. It specifies the nearest endpoint a control can credibly be said to change, the minimum evidence needed, the principal rights burden, and the downstream claim that remains unsupported.

Table A6. Illustrative policy-effect ceilings at specific CDE nodes. Legal authority, implementation quality, rights burdens, and review conditions remain context-dependent. Source: author synthesis of programme and public evidence.

Governance option	CDE node / evidence holder	Nearest measurable endpoint	Minimum credible evidence	Rights and reversal	Unsupported downstream claim
Synthetic political impersonation	Deployment → exposure; sender, telecom, platform, election authority	Verified-origin coverage; confirmed impersonations blocked before delivery; false positives, bypass, and interruption time	Delivery-level records with an auditable denominator and independently adjudicated samples	Protect satire, anonymity, privacy, and appeal; time-limit protocols and restore wrongful blocks	Belief, turnout, vote choice, or election outcome changed
Sensitive or vulnerability data for political targeting	Deployment; campaign, data controller, broker, platform, DPA	Lawful-purpose and provenance coverage; prohibited joins detected; deletion or access requests completed	Scoped data-lineage audit and adversarial sample, not self-attestation alone	Protect privacy, association, organising, and security; deletion deadlines, necessity review, and appeal	Compliant data use prevented manipulation or electoral harm
Tool-enabled generation-to-distribution workflow	Capability → deployment; provider, deployer, tool operator, credential owner	Prohibited-action success; authorisation coverage; escalation; revocation latency; rollback success	Exact configured system, preserved logs and artefacts, matched sandbox controls, and clean replay	Protect security and legitimate automation; graduated permissions, expiry, rollback, and retest after change	Prevalence of misuse, authentic exposure, or human effect
Opaque or high-frequency distribution	Deployment → exposure; platform, advertiser, intermediary, qualified researcher	Deduplicated authentic reach; frequency; targeting accuracy; recommendation or forwarding uplift; correction-window time	Privacy-preserving delivery data with an auditable denominator and appropriate comparison design	Protect privacy, speech, and equal access; post-window review, disparate-burden tests, and reversal	Persuasion, agency transfer, turnout change, or electoral materiality
Weak evidence custody or route opacity	Across CDE bridges; provider, deployer, auditor, regulator, or court	Route-receipt completeness; artefact preservation; hash/schema validity; clean reload; inter-auditor reproduction; documented missingness	Versioned, content-addressed artefacts under independent or escrowed custody with tiered access	Protect privacy, confidentiality, cybersecurity, IP, and proportionality; minimise data and limit retention	The system was safe, manipulative, effective, or electorally consequential
Slow incident interruption and remedy	Exposure → response; platform, telecom, election authority, court or law enforcement by mandate	Detection-to-preservation, attribution, interruption, notice, appeal, and restoration time; evidence-loss and wrongful-action rates	Incident timelines, outcome-coded appeals, and independent after-action review	Protect due process and against over-removal; automatic expiry, prompt review, restoration, and aggregate reporting	Votes were “saved,” belief change was averted, or the intervention worked election-wide

G. Source-linked policy review: findings and claim ceilings

Scope. The 1 September supplement individually annotates the existing 118 implementations and keeps three additional candidates outside that denominator: accuracy-focused sharing prompts, timely claim-specific corrective context, and an independently governed dismissible conversational monitor. Its 69 selected source records generate 278 source–implementation links; overlapping studies, versions, and repeated mappings must not be counted as independent corroboration (Guerrero 2026f).

Depth. The review recorded eight new main-text or substantive-section checks: Slaughter et al. (2025), Biddlestone et al. (2025), Ancina et al. (2026), DeVerna et al. (2024), Debenedetti et al. (2025), Poonsiriwong et al. (2026), Ibrahim et al. (2026), and Salvi et al. (2026). Other source records have explicitly different depths, including publication abstracts or summaries, official records, and six retained prior adjudications. The manuscript integration rechecked key passages and bibliographic metadata; neither exercise independently reproduces the source data or constitutes a double-screened systematic review.

Interpretation. The H/T/O/A/N categories in the supplement indicate kinds of support—human, technical, observational/implementation, adjacent-domain, and normative/proposal—not a strength ranking. A–F remains historical recommendation metadata. Source-specific matches and limitations govern the revised manuscript; the production Atlas’s claim-check flags remain unchanged. A component effect, a law’s existence, and preserved agency require different propositions and evidence.

Table A7. Selected external findings integrated into Part IV. Study findings are separated from the paper’s agency-transfer inferences. The full 118-row source ledger is Guerrero 2026f.

Source and design	Observed or reported result	Ceiling and research implication
Nouwens et al. 2020 Field interface experiment; n = 40	The availability of a first-screen rejection option altered cookie acceptance.	Recorded choice is not informed preference alignment; a statutory ban and its enforcement were not the treatment.
Ershov and Morales 2024 Quasi-experimental Twitter change	Lower news sharing, without a significant selective reduction for lower-factualness outlets; heterogeneous responses.	Generic friction is not accuracy-focused prompting or a WhatsApp cap. Include lost legitimate circulation in the evaluation.
Pennycook et al. 2021; Pennycook and Rand 2022 Surveys, field experiment; internal meta-analysis	Accuracy-oriented attention improved bounded sharing-quality or discernment outcomes.	Source-quality and stated-sharing proxies do not establish agency. The internal meta-analysis is not twenty independent-team replications.
Roozenbeek et al. 2022; Pennycook et al. 2024; Biddlestone et al. 2025; Ancina et al. 2026	Technique recognition, truth discernment and sharing do not necessarily improve together. Some benefits are treatment- or topic-specific.	Assess accurate emotional content, untaught tactics and follow-up. In Ancina et al., n = 154 and simulated messenger exchanges limit transfer.
Slaughter et al. 2025, 2026 Synthetic-control observational analysis	Post-attachment attenuation is larger than the effect relative to the post total including earlier engagement.	The post-note and total windows differ; neither is an all-platform rollout estimate. The correction concerns competing interests, not revised effects.
Ibrahim et al. 2026 Preregistered experiment; analysed n = 2,610; preprint	Warnings shifted some perceptions, without reliable beneficial effects on self-perceived rightness or repair intentions.	Eight-turn conflicts and self-reports; no demonstrated relationship repair or general zero-effect conclusion.
Poonsiriwong et al. 2026 Five-arm experiment; n = 150; preprint	A dismissible monitor improved the secondary AI-steered-choice endpoint, not the primary detection measure.	Thirty participants per arm, simulated tasks; no established advantage over forcing, durable agency outcome, or demonstrated local deployment.

Source and design	Observed or reported result	Ceiling and research implication
Salvi, Cuevas, and Horta Ribeiro 2026 Two experiments; n = 2,012; preprint	Promotional conversation shifted book selections. Disclosure did not significantly attenuate steering in the reported comparison.	Disclosure and baseline span collection waves, not one contemporaneous five-arm randomisation. Selection is not established purchase behaviour or durable agency loss.
DeVerna et al. 2024 Preregistered experiment; n = 2,159	No average discernment improvement from tested LLM fact checks; adverse responses to some erroneous or uncertain outputs.	Forty headlines and one older configuration. Human comparator collected later; optional viewing is selected rather than randomised exposure.
Debenedetti et al. 2025 CaMeL technical evaluation, v2	Control/data-flow separation and permission policies constrain external prompt injection under the stated assumptions.	Initiating query trusted; some text-only influence out of scope. Protecting the principal is not protecting people from that principal.
Turner et al. 2021; Allcott et al. 2025, revised 2026 IoT exercises; search field experiments	Data export did not deliver functional migration in the IoT exercises. Choice, default and trial interventions differ in search.	Historical and adjacent domains, not persistent-assistant migration. Search-market counterfactuals are modelled. Test continuation, not a download alone.
Biega et al. 2020; Fang et al. 2025 Offline recommendation; four-week chatbot study	Data budgets affect recommender quality. The chatbot study distinguishes assigned conditions from voluntary-use associations.	Neither is a randomised test of conversational memory limits. Preserve the distinction between usage association and treatment effect.

Analytical conclusion. Outcome mismatch, incomplete delivery, and protection of the wrong principal are three distinct reasons a component result may not justify a democratic-protection claim. The independently governed monitor-plus-authority-limits-plus-exit package is a plausible hypothesis. Durable reflective agency, institutional self-government, and the package's distributional or concentration effects remain open questions. No new societal effect estimate is claimed.

Source independence and versions. Preprints are identified in text and references; their study status is not upgraded by inclusion here. Relevant industry affiliations, funding and corrections are recorded in the review ledger. Slaughter et al. (2026) adds disclosure concerning prior Meta and X/Twitter collaborations. Such interests warrant scrutiny, not automatic dismissal. The current review is not a complete audit of legal applicability; existing legal sections and the new empirical supplement have separate scopes.

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Source status. Guerrero (2026a–f) are materials produced or curated within this ERA:AI research programme, not independent corroboration. The Part IV empirical results belong to the external studies cited below; the programme's synthesis and governance inferences are labelled separately. Preprints, technical evaluations, internal meta-analysis, observational estimates, and corrections retain their respective status. Inclusion here is not an independent replication. Provider reports remain self-reported evidence unless independently verified.

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ERA:AI project evidence and research materials

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